



STEPCONE-2025

Standard Operating Procedure (SOP)

Name of the Event: CAD-MANIA

Conducted by: Department of Mechanical Engineering

Duration of Event: hours

Expected No of Registrations: 25 teams

Registration Fee: 1000/-

Date of the Event: 24-01-2025(Tentative)

About the Event:

The CAD-MANIA contest is designed to challenge participants to showcase their skills in 3D modelling, assembly, and simulation through a series of tasks that test creativity, technical proficiency, and analytical abilities.

It consists of three rounds.

Details of the Event:

Rounds:

Round 1: Quiz on General Mechanical Concepts

- Teams will answer a series of questions on general mechanical basic concepts.
- A time limit will be imposed for each question to ensure quick responses.
- Evaluation Criteria: Number of correct answers within the time limit.

Round 2: Problem Allocation for Assembly Drawing

- Teams will be assigned a problem requiring the creation of an assembly drawing using any 3D modeling software of their choice.
- Evaluation Criteria: Adherence to design specifications, accuracy, and completeness of the assembly drawing.

Round 3: Structural and Thermal Analysis

- A common problem will be allocated to all teams. Participants must perform structural and thermal analysis of the given assembly using analysis software of their choice.
- Evaluation Criteria: Accuracy of simulations, insights drawn from the results, and proposed design improvements.

Venue	Duration	Faculty Coordinator	Student Coordinator
CAD-LAB Dept of Mechanical	Hours	Mr. G. Sasidhar Dr. Bappa Mondal	G .Asritha Vatsala P. Manikanta L. Bhuvan Manikar



Guidelines of the Event:

- Teams may consist of up to 3 participants. Individual participation is allowed.
- Participants must bring their laptops pre-installed with 3D modeling and simulation software of their choice.
- Each team must adhere to the submission deadlines for each round.
- Use of unlicensed or pirated software is strictly prohibited.
- Participants must ensure the originality of their designs. Plagiarism will result in disqualification.
- Internet access will be provided for research purposes but not for file sharing or external collaborations.
- Coordinators reserve the right to disqualify teams found violating any guidelines.

Evaluation and result:

- Judges will evaluate participants based on their performance in all three rounds.
- The final results will be announced after consolidating scores from all three rounds.
- Total Points = $p(\text{Round 1}) + p(\text{Round 2}) + p(\text{Round 3})$.
- Judges' decisions will be final and binding.